Probability of Single Events

Name _____ Date _____ No calculator Version 1

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KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · equally likely *(tūpono ōrite)* · expected number

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 7 on an ordinary dice _____

b) Getting a head when a coin is tossed _____

c) Rolling a number less than 7 on a dice _____

d) Rolling a 6 on a dice _____

e) Picking a red counter from a bag of 9 red and 1 blue _____

f) Rolling an even number on a dice _____

g) Picking a red card from a normal pack of 52 cards _____

h) Rolling a number less than 6 on a dice _____

B Equally likely outcomes ●○○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 5 **b)** an even number **c)** a number greater than 4

d) a prime number **e)** a number less than 7 **f)** an 8

3. A bag holds 5 red, 3 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

e) red or blue **f)** not red **g)** green **h)** red, blue or green

4. Cards numbered 1 to 20 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

e) an odd number **f)** a multiple of 4 **g)** less than 6 **h)** a two-digit number

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C The probability of not happening ●●○

Remember: An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.3 **b)** $\frac{2}{5}$ **c)** 35%

d) 0.82 **e)** $\frac{7}{12}$ **f)** 4%

g) 0.55 **h)** $\frac{1}{8}$ **i)** 62%

D One event or another ●●○

Remember: If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 2 or a 5 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 5 red, 3 blue and 2 green counters

d) a number less than 3 or greater than 15 from cards numbered 1 to 20

e) a 1, a 2 or a 3 on a dice

f) a prime number or a 6 on a dice

E Expected number ●●○

Remember: Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 120 times. How many sixes would you expect?

b) A coin is tossed 250 times. How many heads would you expect?

c) The probability that a spinner lands on red is 0.35. It is spun 200 times. How many reds would you expect?

d) The probability that a bus is late is 0.08. How many late buses would you expect in 50 days?

e) A bag holds 3 blue and 5 white counters. A counter is picked and put back 400 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{20}$. How many prizes would you expect from 300 tickets?

g) A spinner has 5 equal sections and 2 of them are green. It is spun 150 times. How many greens would you expect?

h) The probability that a bus is on time is 0.9. How many of the next 80 buses would you expect to be on time?

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F Problems ●●●

8. A letter is picked at random from the word MATHEMATICS. Find the probability that it is the letter T.

9. A raffle has 200 tickets and Mia buys 8 of them. Find the probability that Mia wins, as a fraction in its simplest form.

10. The probability that a seed grows is 0.85. If 60 seeds are planted, how many would you expect to grow?

11. A box holds 24 chocolates. The probability of picking a dark chocolate is $\frac{3}{8}$. How many dark chocolates are in the box?

12. Three runners are in a race. The probability that Ana wins is 0.25 and the probability that Ben wins is 0.4. Find the probability that the third runner, Caleb, wins.

13. A bag holds 4 red counters, 6 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

14. A spinner has 5 equal sections, and 2 of them are red. It is spun 80 times. How many times would you expect it not to land on red?

★ Extension

15. Bag A holds 8 counters and 3 of them are red. Bag B holds 12 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

16. A dice is rolled. Find the probability that the number is prime or even (or both).

17. A number is picked at random from 1 to 30. Find the probability that it is a multiple of 3 and also a multiple of 5.

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Probability of Single Events

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KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · equally likely *(tūpono ōrite)* · expected number

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 0 on an ordinary dice _____

b) Getting a tail when a coin is tossed _____

c) Rolling a number greater than 0 on a dice _____

d) Rolling a 1 on a dice _____

e) Picking a blue counter from a bag of 8 blue and 2 red _____

f) Rolling an odd number on a dice _____

g) Picking a black card from a normal pack of 52 cards _____

h) Rolling a number greater than 1 on a dice _____

B Equally likely outcomes ●○○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 2 **b)** an odd number **c)** a number less than 3

d) a multiple of 3 **e)** a number greater than 0 **f)** a 9

3. A bag holds 4 red, 6 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

e) red or blue **f)** not red **g)** green **h)** red, blue or green

4. Cards numbered 1 to 24 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

e) an odd number **f)** a multiple of 4 **g)** less than 6 **h)** a two-digit number

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C The probability of not happening ●●○

Remember: An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.4 **b)** $\frac{3}{7}$ **c)** 45%

d) 0.76 **e)** $\frac{5}{12}$ **f)** 8%

g) 0.15 **h)** $\frac{3}{10}$ **i)** 27%

D One event or another ●●○

Remember: If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 3 or a 4 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 4 red, 6 blue and 2 green counters

d) a number less than 4 or greater than 16 from cards numbered 1 to 20

e) a 4, a 5 or a 6 on a dice

f) a square number or a 5 on a dice

E Expected number ●●○

Remember: Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 180 times. How many fours would you expect?

b) A coin is tossed 350 times. How many tails would you expect?

c) The probability that a spinner lands on blue is 0.45. It is spun 200 times. How many blues would you expect?

d) The probability that a train is late is 0.06. How many late trains would you expect in 50 days?

e) A bag holds 5 blue and 3 white counters. A counter is picked and put back 240 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{25}$. How many prizes would you expect from 400 tickets?

g) A spinner has 5 equal sections and 3 of them are green. It is spun 150 times. How many greens would you expect?

h) The probability that a bus is on time is 0.85. How many of the next 80 buses would you expect to be on time?

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F Problems ●●●

8. A letter is picked at random from the word STATISTICS. Find the probability that it is the letter S.

9. A raffle has 250 tickets and Leo buys 10 of them. Find the probability that Leo wins, as a fraction in its simplest form.

10. The probability that a seed grows is 0.9. If 40 seeds are planted, how many would you expect to grow?

11. A box holds 30 chocolates. The probability of picking a dark chocolate is $\frac{2}{5}$. How many dark chocolates are in the box?

12. Three runners are in a race. The probability that Ana wins is 0.3 and the probability that Ben wins is 0.45. Find the probability that the third runner, Caleb, wins.

13. A bag holds 5 red counters, 7 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

14. A spinner has 8 equal sections, and 3 of them are red. It is spun 120 times. How many times would you expect it not to land on red?

★ Extension

15. Bag A holds 10 counters and 4 of them are red. Bag B holds 14 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

16. A dice is rolled. Find the probability that the number is odd or greater than 4 (or both).

17. A number is picked at random from 1 to 40. Find the probability that it is a multiple of 4 and also a multiple of 6.

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